

# Prisha Singla

AI Engineer - AI Agents · RAG Pipelines · LLM Systems · Production LLM Systems

+91 9911527990 | [prishasingla23@gmail.com](mailto:prishasingla23@gmail.com) | [LinkedIn](#) | [GitHub: Prisha-singla-dev](#) | Bengaluru, India

## PROFESSIONAL SUMMARY

AI Engineer with 2.5 years of experience building production conversational AI, agentic AI workflows, and LLM-powered systems at Capgemini, serving 250+ daily users. Shipped systems that reduced operational overhead by 40%, improved NL2SQL accuracy by 20 percentage points, cut LLM spend by 12-15%, and reduced MTTR by 50%. Strong in Python, Google ADK, LangGraph, prompt engineering, retrieval systems, production deployment, and cloud-based AI services.

## WORK EXPERIENCE

**AI Engineer (Senior Software Engineer Analyst) | Capgemini Technology Services, Bengaluru** **December 2023 – current**

- Architected and shipped production AI agent workflows using LangGraph and Google ADK on AWS ECS and Docker, serving 250+ daily users and reducing operational overhead by 40%.
- Improved production NL2SQL accuracy by 20 percentage points by implementing schema linking, structured validation, retrieval-aware response generation, and prompt optimization.
- Eliminated 35% of repetitive analyst tasks by designing execution agents with intent interpretation, planning, human-in-the-loop approvals, and safe database write workflows.
- Built monitoring, evaluation, and failure-triage loops for deployed AI systems using CloudWatch dashboards, log correlation, and distributed tracing, reducing MTTR by 50%.
- Reduced LLM API spend by 12-15% by implementing prompt compression, intelligent caching, and ML-based model routing as a reusable optimization module for production AI services.
- Designed governed data-access workflows with audit trails and controlled read/write operations, reducing rework incidents by 38% while improving traceability, safety, and production reliability.

## PROJECTS

**CodeSentinel | LangGraph · FastAPI · React/Vite · Gemini → OpenRouter · Render · Vercel | [GitHub](#) | [Live Demo](#) June 2026**

- Developed a production multi-agent AI code review platform with four specialized LangGraph agents that analyze pull requests for security vulnerabilities, performance bottlenecks, logical defects, and code quality, enabling automated GitHub review workflows.
- Built reliability and regression-resistant agent behavior through structured JSON enforcement, response repair, cascading fallback across seven LLM providers, and resilient API handling, eliminating silent failures caused by malformed outputs, rate limits, and model outages.

**LLM Cost Firewall | Python · FastAPI · Gemini API · scikit-learn · GitHub Actions · Render | [GitHub](#) | [Live Demo](#) April 2026**

- Reduced average LLM inference cost by 60% through a production API gateway that dynamically routed requests between Gemini 2.5 Flash and Gemini 2.5 Pro using a six-factor ML complexity scoring model.
- Eliminated redundant LLM requests using a semantic retrieval layer combining SHA-256 exact matching with embedding-based similarity search, improving knowledge reuse and response efficiency.
- Improved production reliability by implementing rate limiting, budget enforcement, exponential backoff, 37 automated unit and integration tests, and GitHub Actions CI/CD, creating a maintainable and production-ready AI service.

## TECHNICAL SKILLS

- **AI / LLM:** LangGraph, LangChain, CrewAI, Google ADK, RAG Pipelines, NL2SQL, Prompt Engineering, Embeddings, Vector Search, LLM Evaluation, Observability
- **Production AI:** AI Agents, Evaluation Loops, Failure Analysis, Structured Validation, Human-in-the-loop Workflows, Model Routing, Audit Trails
- **Cloud / Infra:** AWS, Azure AI Foundry, Docker, Kubernetes, GitHub Actions CI/CD, API Gateway, CloudWatch
- **Backend / Data:** Python, FastAPI, PostgreSQL, Redis, MongoDB, REST APIs, WebSockets, Query Optimization
- **ML / Supporting Tools:** scikit-learn, Semantic Caching, Cost Optimization, Distributed Tracing

## CERTIFICATIONS & RECOGNITION

- **AWS AI Practitioner (AIF -C01)**
- **Microsoft Azure AI Engineer Associate (AI – 102)**
- **Google Generative AI Leader**

## EDUCATION

**B. Tech, Computer Science Engineering · 8.8 CGPA | Graphic Era Hill University, Dehradun**

**August 2019 – July 2023**